

CHAPTER 18

PROGRAM DESIGN FOR RESISTANCE TRAINING

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After completing this chapter, you will be able to

- evaluate the requirements and characteristics of a sport and assess an athlete for the purpose of designing a resistance training program;
- select exercises based on type, specificity, technique experience, equipment availability, and time availability;
- determine training frequency based on training status, sport season (i.e., phase of the season), load, exercise type, and other concurrent exercise;
- arrange exercises in a training session according to their type;
- determine 1-repetition maximum (1RM), predicted 1RM from a multiple RM, and RM loads;
- assign load and repetitions based on the training goal;
- know when and by how much an exercise load should be increased;
- assign training volumes according to the athlete's training status and the training goal; and
- determine rest period lengths based on the training goal.

Effective training programs involve the coordination of many variables in a sequential and systematic fashion that enables the body to appropriately adapt to specific exercise stimuli and subsequently improve performance. Having a basic understanding of the physiological responses to various training stimuli is essential for the strength and conditioning professional to be able to coordinate the various training aspects successfully. When focusing on the resistance

training component of a comprehensive training program, it is helpful to approach the task one program element at a time, keeping in mind the primary principles of anaerobic exercise prescription.

Principles of Anaerobic Exercise Prescription

Resistance training programs for athletic populations require attention to the principles of specificity, overload, progression, variation, reversibility, and individualization. One of the most basic concepts to incorporate in all training programs is **specificity**. The term, first suggested by DeLorme in 1945 (39), refers to the method whereby an athlete is trained in a specific manner to produce a specific adaptation or training outcome. In the case of resistance training, specificity refers to aspects such as the muscles involved, the muscle action (e.g., concentric, eccentric, and isometric), and the movement velocity and coordination (i.e., movement pattern; this does not mean replicating specific sporting movements), but it does not always reflect the combination of all these aspects. Importantly, it does not mean that all aspects of the training must directly mimic that of the sporting skill. For example, a squat movement is relevant to the vertical jump because it involves overcoming resistance in a similar movement pattern (e.g., combined knee and hip flexion followed by forceful knee and hip extension) and engages muscles (i.e., predominantly the knee and hip extensors) that are involved in the vertical jump, yet the speed of movement and force application are disparate between the squat and vertical jump. Sometimes used interchangeably with *specificity* is the acronym **SAID**, which stands for *specific adaptation to imposed demands*. The underlying principle is that the type of demand placed on the body dictates the type of adaptation that will occur. For instance, athletes training for power in high-speed movements (e.g., baseball pitch or tennis serve) should attempt to activate or recruit the same motor units required by their sport at the highest velocity possible (160, 163). Additionally, specific improvements in force and velocity characteristics may occur following heavy or ballistic resistance training, with greater improvements in power production occurring in response to strength training targeting increases in maximal strength in weak athletes (26, 28, 29). Specificity also relates to the athlete's sport season. As an athlete progresses through the preseason, in-season, and postseason, all forms of training should gradually progress in an organized manner from general to specific (159). Although participation in the sport itself provides the greatest opportunity to improve performance in the sport, the proper application of the specificity principle certainly increases the likelihood that other training will also positively contribute to performance via the transference of training adaptations in the gym to the sport-specific skills.

Overload refers to the application of a training stimulus that exceeds what the athlete is accustomed to (i.e., increased load, sets, or repetitions [i.e., volume]) and produces an adaptation beyond the baseline abilities of the

athlete. Without an overload stimulus, even an otherwise well-designed program greatly limits an athlete's ability to make improvements. The obvious application of this principle in the design of resistance training programs involves increasing the loads assigned in the exercises, especially if the primary training goal is to improve maximal strength. Other more subtle changes include altering the number of training sessions per week (or per day in some instances) or increasing training **volume** by adding exercises or sets, which may preferentially stimulate a hypertrophic response (96, 143). Alternatively, varying exercise choice, modifying rest periods between sets and exercises, or any combination of these can result in an overload. The basis of the overload principle is to provide the body with a unique training stimulus to facilitate the development of the desired adaptations. Although an overload stimulus is typically thought of as an increase of a given training variable (e.g., load, volume, **velocity**), a reduction in these variables may also provide the opportunity for adaptation to take place. For example, a reduction in intensity (i.e., relative load) allows for an increase in the speed of a movement, thus facilitating the development of rapid force production characteristics (i.e., speed–strength). Alternatively, a reduction in training volume while maintaining (or increasing) intensity during a deload week may lead to an enhancement in performance ([figure 18.1](#)) if a sufficient stimulus (i.e., volume and intensity) has been applied previously. When the overload principle is properly and systematically applied, and all other components of training are considered (e.g., skill-based technical and tactical training, conditioning, and external stressors), overtraining can be avoided and the desired training adaptation will occur.

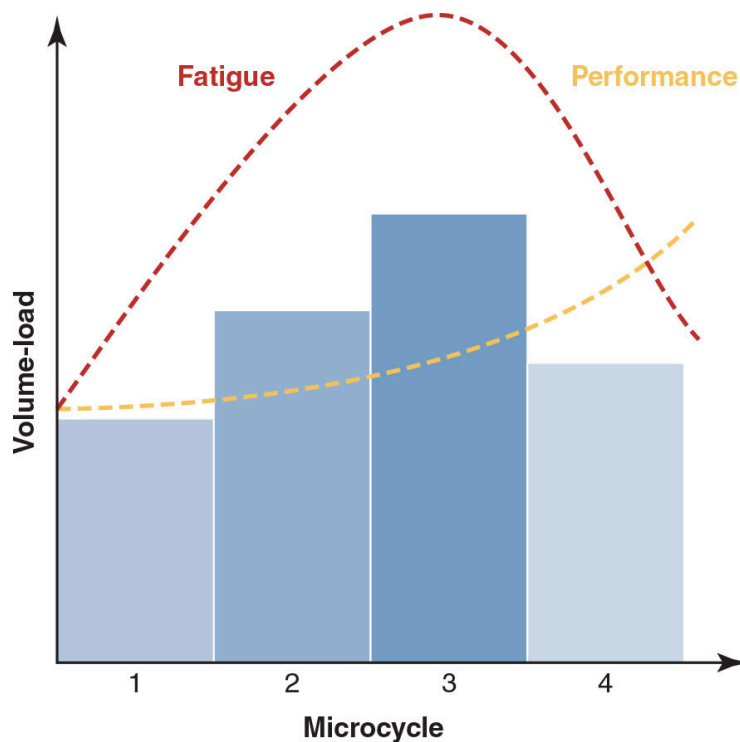


FIGURE 18.1 Example of changes in fatigue and performance (i.e., fitness-fatigue paradigm) across a mesocycle with a deload week. As the volume load increases across the first three weeks; some accumulation of fatigue occurs. Only once there is a reduction in volume load, during the deload week, does the increased performance manifest.

If a training program is to continue producing higher levels of performance, the intensity or volume (i.e., overload) of training must continue to provide a unique stimulus. **Progression**, when applied properly, with loading or volume being sequential and specific to the desired adaptive response, promotes long-term training benefits. Although it is customary to focus only on the resistance used (i.e., intensity), one can progressively increase training volume by raising the number of weekly training sessions, adding more drills or exercises to each session, increasing the number of sets performed, or changing the type or technical requirements of the drills or exercises. For example, an athlete may progress from the front squat and clean grip shoulder shrug to learning the hang power clean and eventually the power clean through the use of a sequential application of various training exercises that segment the key positions associated with the lift (21). A key factor governing how training progression is applied is that it must be based on the athlete's training needs and training status and it needs to be introduced systematically and gradually. It is also worth noting that progressively increasing volume tends to emphasize hypertrophic adaptations, whereas increasing intensity emphasizes strength-related adaptations (35, 97, 125, 144).

Variation within resistance training programs focuses on removing linearity by manipulating overload and specificity to promote the desired adaptations. For example, strength and conditioning professionals may introduce variation into a training program by modifying volume, intensity, set configuration, rest periods, and exercise selection to provide a novel overload stimulus. These modifications to a training program may help reduce training monotony while also potentially reducing fatigue and providing a range of training stimuli. Regarding training intensity, strength–power characteristics have been developed within training programs that include more variation compared to prescribing constant repetition ranges with a greater or fewer number of sets and training or not training to failure (121, 195, 196). Moreover, it has been suggested that maximal and rapid force production may be improved to a greater extent using a combination of heavy and light training intensities compared to performing only one loading method (68, 118, 189).

Strength and conditioning professionals can add variation to a training program at both the macro and micro level in several ways. At the macro level, implementing a periodized training plan may help develop specific fitness characteristics (e.g., hypertrophy, strength, and power) at different times throughout the training year. This in turn may provide athletes with a variety of training stimuli and potentially enhance the adaptations in subsequent training phases. Within each phase of training strength and conditioning professionals can make micro-level adjustments and add variation by modifying training intensities and exercise selection, thereby reducing training monotony and offering unique training stimuli.

As mentioned previously, the introduction of an overload stimulus is meant to promote the development of a desired fitness characteristic. However, the removal or reduction of a training stimulus or the involution from monotonous training and poor fatigue management may cause the opposite to occur; this deterioration or loss of fitness characteristics refers to the principle of **reversibility**. An example of the deterioration of a fitness characteristic after the removal of a training stimulus is the reduction in rapid force production that may occur during a higher volume phase focused on improving muscle hypertrophy (98, 164). However, the opposite may also occur because a loss in muscle size (i.e., muscle atrophy) may occur when training volume is reduced during a taper (7, 188). As such, reversibility may be considered a product of programming variation because different fitness characteristics are emphasized during various times throughout the training year. Regarding involution, a plateau or decreased performance may result because of a lack of mechanical variation (52) or appropriate loading (176). For example, maximal strength adaptations may plateau if a similar training program with small progressions in

loading is prescribed for longer than six weeks (186). Involution related to poor fatigue management may be the product of nonfunctional overreaching without the inclusion of proper recovery, however (54, 55). While the loss of fitness characteristics could be mitigated by incorporating retaining loads (i.e., small percentage of overall training volume focused on maintaining a desired characteristic) (42), it is important that strength and conditioning professionals understand the difference between retaining loads and the minimal effective dose needed to maintain or develop a fitness characteristic.

The training goals and needs of each athlete are unique and should therefore command a training program suited specifically for them; this training principle is termed **individualization**. Athletes competing in different sports, the same sport, or in different events come with different genetic characteristics and training histories that affect how they respond to various training stimuli. As such, strength and conditioning professionals should always avoid a cookie cutter approach to designing training programs and should instead focus on providing individualized training programs that help each athlete reach their training goals. While this may seem simplistic, it is important to note that many aspects of different training programs may be similar or even identical. For example, general strength characteristics for the lower body may be addressed by prescribing one of many core exercises such as the back squat, leg press, hex-bar deadlift, and so on ([figure 18.2](#)); however, a training program focused on this fitness characteristic will likely have more than one lower body exercise, and thus the likelihood of overlap between programs is high. Taking this into consideration, strength and conditioning professionals may find that 80% to 90% of the exercises used in a training program are similar for multiple athletes, while 10% to 20% of the exercises will be individualized. It is important to note that the loads programmed for these exercises are always individualized to each athlete's capabilities.

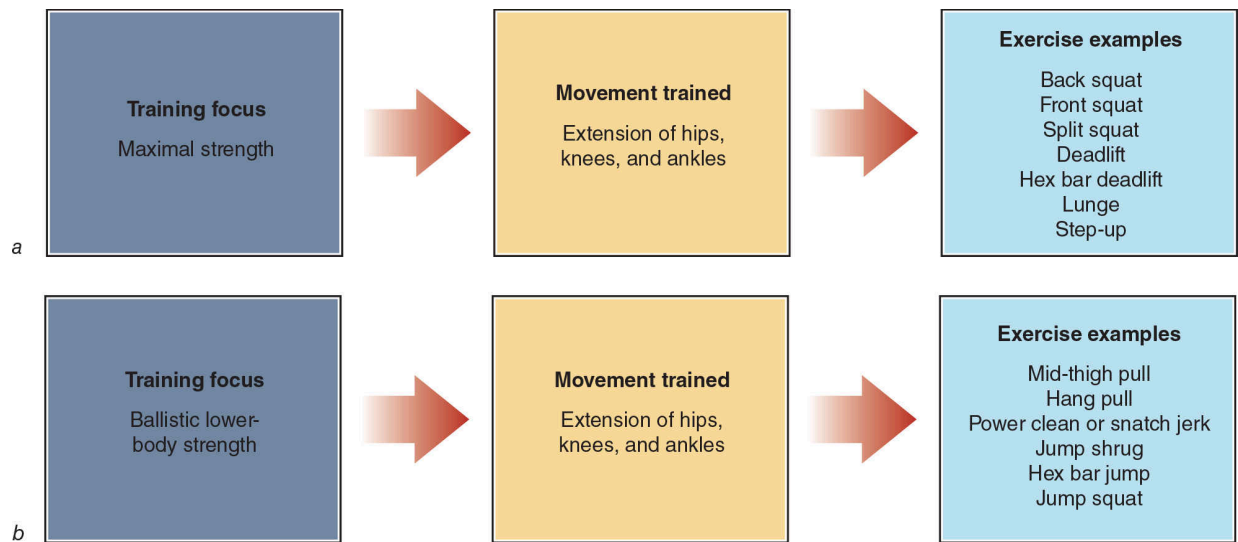


FIGURE 18.2 Exercise selection flow chart for lower body (a) strength and (b) power. *Note:* The displayed exercises are examples and do not comprise an exhaustive list.

Designing a resistance training program is a complex process that requires the recognition and manipulation of seven **program design** variables (referred to in this chapter as [steps 1](#) through [7](#)).

This chapter discusses each variable in the context of three scenarios that enable the strength and conditioning professional to see how training principles and program design guidelines can be integrated into an overall program.

1. Needs analysis
2. Exercise selection
3. Training frequency
4. Exercise order
5. Training load and repetitions
6. Volume
7. Rest periods

Athlete Scenarios

The three scenarios include a basketball center (scenario A) in her in-season, an American football offensive lineman (scenario B) during his off-season, and a cross country runner (scenario C) during his preseason. It is understood that in each scenario, the athlete is well conditioned for his or her sport, has no musculoskeletal dysfunction, and has been cleared for training and competition by the sports medicine staff. The athletes in scenarios A (basketball center) and B (American football lineman) have been resistance training since high school, are accustomed to lifting heavy loads, and are skilled in machine and free weight exercises. The high school cross country runner in scenario C, in contrast, began a resistance training program in the preseason only four weeks ago, so his training is limited, and his exercise technique skills are not well developed.

Scenario A		Scenario B		Scenario C	
Sex	Female	Sex	Male	Sex	Male
Age	20 years old	Age	28 years old	Age	17 years old
Sport	Collegiate basketball	Sport	Professional American football	Sport	High school cross country running
Position	Center	Position	Offensive lineman	Position	(Not applicable)
Season	Beginning of the in-season	Season	Beginning of the off-season	Season	Beginning of the preseason

Step 1: Needs Analysis

The strength and conditioning professional's initial task is to perform a **needs analysis**, a two-stage process that includes an evaluation of the requirements and characteristics of the sport and an assessment of the athlete.

Evaluation of the Sport

The first task in a needs analysis is to determine the unique characteristics of the sport, which includes the general physiological and biomechanical profile, common injuries and mechanisms of injury, and position-specific attributes. This information enables the strength and conditioning professional to design a program specific to those requirements and characteristics. Although this task can be approached in several ways, it should at least include consideration of the following attributes of the sport (50, 128):

- Body and limb movement patterns and muscular involvement (**movement analysis**)
- Strength, power, hypertrophy, and muscular endurance priorities (**physiological analysis**)
- Common sites for joint and muscle injury and causative factors (**injury analysis**), especially for non-contact injuries

Other characteristics of a sport—such as cardiovascular endurance, speed, agility, and flexibility requirements—should also be evaluated. This chapter, however, focuses only on the physiological outcomes that specifically relate to resistance training program design: strength, power, hypertrophy, and muscular endurance.

For example, a movement analysis of the shot-put field event reveals that it is an all-body movement that begins with the athlete in a semi-crouched stance, with many joints flexed and adducted, and culminates in an upright stance with many joints extended and abducted. The most heavily recruited muscles (not in order) are the elbow extensors (triceps brachii), shoulder abductors (deltoids), hip extensors (gluteals, hamstrings), knee extensors (quadriceps), and ankle plantar flexors (soleus, gastrocnemius). Physiologically, shot putting requires high levels of strength and power for a successful performance. Also, enhanced muscular hypertrophy may be advantageous since the muscle's potential to produce force increases as its cross-sectional area becomes greater (81, 102). The muscular endurance requirement is minimal, however. Due to the repetitive nature of training and competition, the muscles and tendons surrounding the shoulder and elbow joints tend to be injured due to overuse (192).

Alternatively, increased hypertrophy may be beneficial for force production, as well as for collision and contact-based sports (e.g., American football or rugby) as a protective mechanism against impacts, and to increase the athletes' momentum (momentum = mass × velocity) to improve outcomes in collisions. For example, if any reduction in speed is disproportionately lower than the increase in muscle mass, the collision sport athlete will be able to generate greater momentum, likely making him or her more difficult to stop or tackle. In contrast, excessive hypertrophy may negatively affect the performance of endurance-based athletes due to the requirement to carry extra mass for longer durations (13). Despite being in the same sport (i.e., athletics) as a shot-put thrower, a distance athlete requires a substantial amount of cardiovascular and muscle endurance within his or her events while also being prone to different overuse injuries (e.g., stress fractures).

Moreover, the amount of hypertrophy may be limited when considering weight-categorized sports such as weightlifting, powerlifting, and combat sports. Depending on the athlete's body composition and body mass, there may only be a limited potential for increases in lean body mass to ensure he or she does not exceed the weight category.

Assessment of the Athlete

The second task is to **profile** the athlete's needs and goals by evaluating training (and injury) status, conducting a variety of tests (e.g., maximum strength testing), evaluating the results, and determining the primary goal of training. The more individualized the assessment process, the more specific the resistance training program for each athlete can be.

Training Status

An athlete's current condition or level of preparedness to begin a new or revised program (**training status**) is an important consideration in the design of training programs. This includes an evaluation by a sports medicine professional of any current or previous injuries that may affect training. Also important is the athlete's **training background** or **exercise history** (training that occurred *before* he or she began a new or revised program), because this information will help the strength and conditioning professional better understand the athlete's training capabilities. Readers are referred to McKay and colleagues (108) for classification of different levels of athlete performance. An assessment of the athlete's training background should examine the

- type of training program (sprint, plyometric, resistance, and so on),
- length of recent regular participation in previous training programs,
- level of intensity involved in previous training programs, and

- degree of **exercise technique experience** (i.e., the knowledge and skill to perform resistance training exercises properly).

[Table 18.1](#) provides an example of how such information might be used to classify athletes' training status as beginner, intermediate, or advanced. The strength and conditioning professional should realize that the three classifications exist on a continuum and cannot be definitively demarcated. For example, in terms of resistance training, many European soccer athletes will have undertaken resistance training across their careers, but this is generally emphasized during preseason training with a relatively low frequency and low training stress, which is further reduced (or completely omitted) in-season, resulting in few of these athletes ever being classified as advanced in terms of resistance training status.

TABLE 18.1 Example of Classifying Resistance Training Status

Resistance training status	Resistance training background				
	Current program	Training age	Frequency (per week)	Training stress*	Technique experience and skill
Beginner (untrained)	Not training or just began training	<2 months	≤1-2	None or low	None or minimal
Intermediate (moderately resistance trained)	Currently training	2-6 months	≤2-3	Medium	Basic
Advanced (well resistance trained)	Currently training	≥1 year	≥3-4	High	High

*In this example, "training stress" refers to the degree of physical demand or stimulus of the resistance training program.

Physical Testing and Evaluation

Physical evaluation involves conducting assessments of the athlete's strength, flexibility, power, speed, muscular endurance, body composition, cardiovascular endurance, and so on. In this chapter, the needs analysis focuses on assessing maximal muscular strength, but a comprehensive assessment goes beyond that. While 1RM testing provides valuable information about the maximal load an athlete can lift during a specific exercise, which aids in programming specific loads, it is important to evaluate other aspects of an athlete's force production ability. On its own, 1RM testing does not provide insight into an athlete's maximal or rapid force-generating capacity. As such, isometric strength assessments, such as the isometric mid-thigh pull, can provide such insight,

especially if combined with assessment of slow stretch-shortening cycle (SSC) function (e.g., countermovement jump) and fast SSC function (e.g., rebound jump) using force plates. Readers are referred to [chapters 13](#) and [14](#) for more information on athlete testing.

In the soccer athlete example provided in [figure 18.3](#), we see that the athlete's peak force and relative peak force during the isometric mid-thigh pull are average for the performances across the league in which he or she competes, highlighting a potential area for development. In contrast, the athlete's countermovement jump height (45 cm) is very good compared to the rest of the league; however, their modified reactive strength index (0.46 [$\text{mRSI} = \text{jump height} / \text{time to takeoff}$]) is only average, due to a prolonged time to takeoff (1.0 s) ([figure 18.4](#)). This indicates that the athlete generated a sufficient impulse (i.e., $\text{force} \times \text{time}$) to achieve the 45-centimeter jump height, using a greater time for acceleration and therefore a suboptimal force profile. As such, the athlete needs to emphasize strength development; as Cormie and colleagues (26, 28, 29) previously demonstrated, strength training and the associated increase in maximal strength result in enhanced force production characteristics during jumping, along with reductions in time to takeoff. Finally, the athlete's rebound jump performance highlights a good ground contact time (163 ms) but a low rebound jump height (22 cm), resulting in an average reactive strength index (1.40) compared to the performances across the league ([figure 18.5](#)). The low jump height may be due to the inability to produce high enough forces during the short ground contact time to provide a sufficient impulse for a higher jump height. Overall, these characteristics highlight that the athlete needs to enhance maximal and rapid force production. As such, initially emphasizing maximal strength for a training block, followed by a subsequent block of training emphasizing speed–strength, may be advantageous.

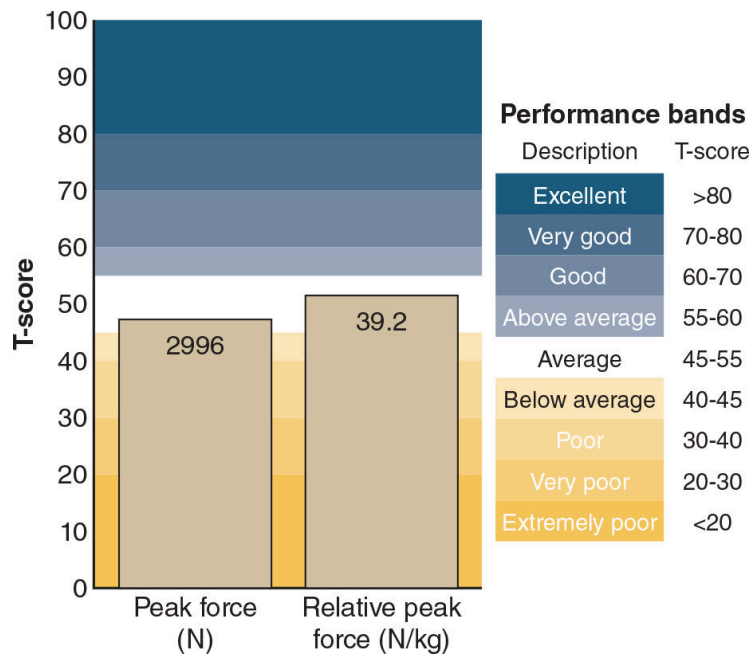


FIGURE 18.3 Individual isometric mid-thigh pull metrics, presented as T-scores for the league in which the athlete competes. Relative peak force (N/kg) is the athlete's peak force (N) divided by body mass (kg).

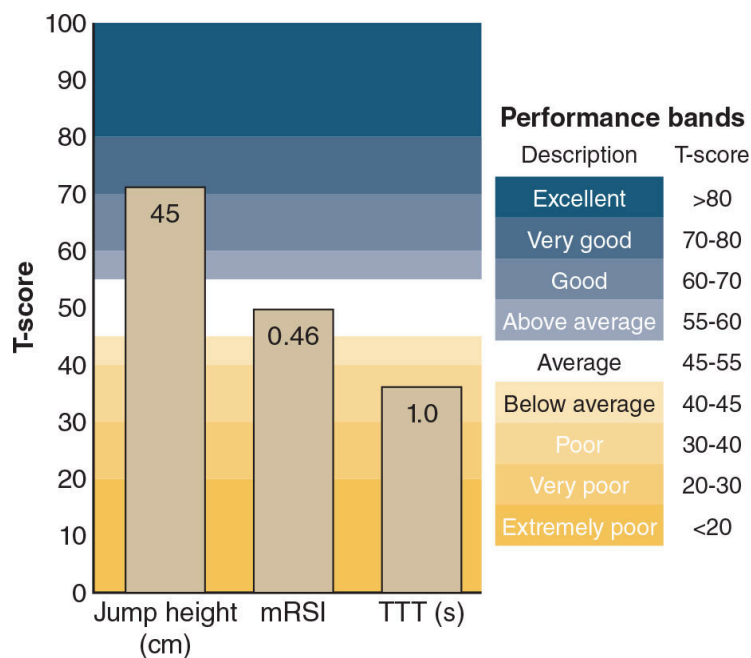


FIGURE 18.4 Individual countermovement jump performance metrics, presented as T-scores for the league in which the athlete competes. mRSI = modified reactive strength index (mRSI = jump height / TTT); TTT = time to takeoff (s).

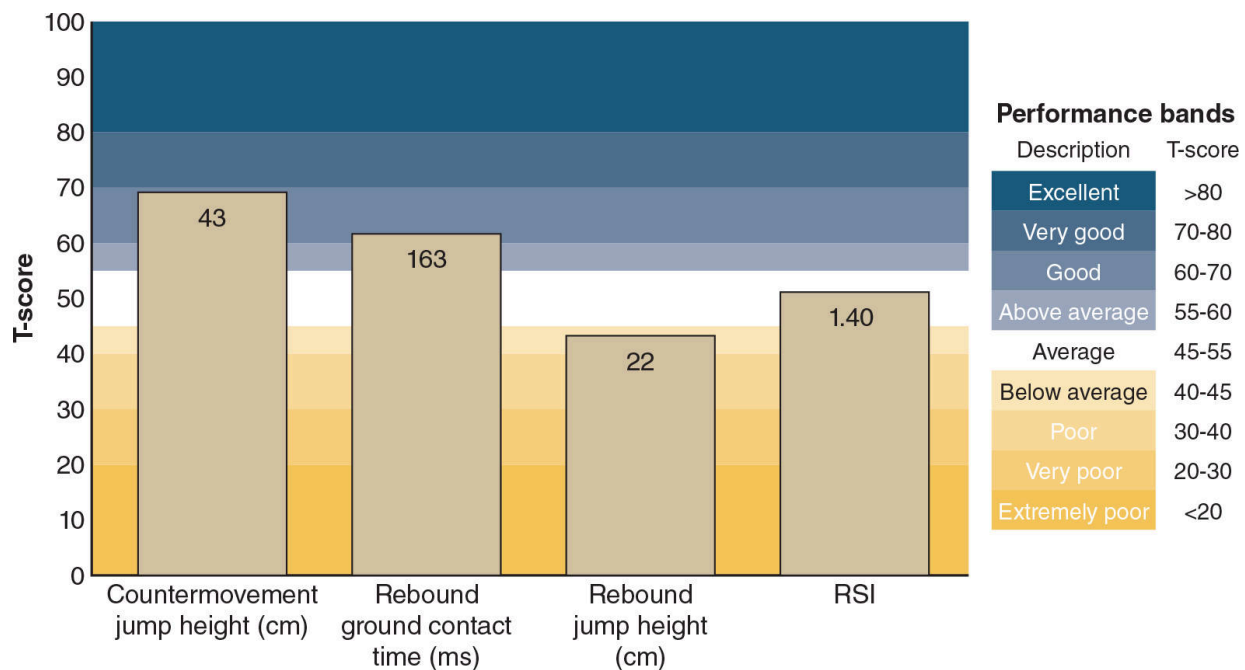


FIGURE 18.5 Individual countermovement rebound jump performance metrics, presented as T-scores for the league in which the athlete competes. RSI = reactive strength index (RSI = rebound jump height / ground contact time).

To yield pertinent and reliable data that can be used effectively to develop a resistance training program, the tests selected should be related to the athlete's sport, consistent with the athlete's level of skill, and realistically based on the equipment available. The result of the movement analysis discussed previously provides direction in selecting tests. Typically, major upper body exercises (e.g., bench press and shoulder press) and exercises that mimic jumping movements to varying degrees (e.g., power clean, squat, front squat) are used in testing batteries. Such assessments should not replace the assessment of jump performance but instead should be used to determine whether any substandard jump performances are due to poor strength levels.

After testing is completed, the results should be compared with normative or descriptive data (if published data exist or at least to squad data in team sports) to determine the athlete's strengths and weaknesses. Based on this evaluation and the needs analysis of the sport, a training program can be developed to improve deficiencies, maintain strengths, or further develop physiological qualities that will enable the athlete to better meet the demands of the sport.

Primary Resistance Training Goal

The athlete's testing results, the movement and physiological analysis of the sport, and the priorities of the athlete's sport season determine the primary goal or outcome for the resistance training program. Typically, this goal is to improve

strength, power, hypertrophy, or muscular endurance. Despite a potential desire or need to make improvements in two different areas (e.g., strength *and* muscular endurance), an effort should be made to concentrate on only one training outcome per training block. This can be accomplished using an emphasis/de-emphasis approach wherein specific fitness characteristics are targeted during different training periods while others are de-emphasized (67). An example of how the strength and conditioning professional may prioritize the resistance training emphases during the four main sport seasons is shown in [table 18.2](#). However, while focusing on the development of a specific physical characteristic, it is important to provide sufficient stimuli for the maintenance of other physical characteristics, especially if they are to be emphasized in subsequent blocks of training. A common strategy in many team sports is to alternate between strength (i.e., strength–speed) and power (i.e., speed–strength) development in-season, because the higher volumes associated with strength endurance– and hypertrophy-focused training can induce excessive fatigue and potentially result in an increased injury risk. However, while this may be easier to accomplish with individual sports or events (e.g., track and field), a team-based sport (e.g., basketball, soccer, hockey) that requires several fitness characteristics for success may have to use a multitargeted approach within training to develop those multiple characteristics simultaneously (177). Within this approach, it has been suggested that compatible fitness characteristics (e.g., strength–power and speed) be trained simultaneously while avoiding incompatible characteristics (e.g., speed and cardiovascular endurance). An emphasis/de-emphasis approach can still be used in these situations (see [chapter 22](#) for more information).

TABLE 18.2 Example of General Training Priorities by Sport Season

Sport season	Priority given to		Resistance training goal*
	Sport practice	Resistance training	
Off-season	Low	High	Hypertrophy and muscular endurance (initially); strength and power (later)
Preseason	Medium	Medium	Sport and movement specific (i.e., strength, power, or muscular endurance, depending on the sport)
In-season	High	Low	Maintenance of preseason training goal; if in-season is long, these goals may alternate between different characteristics (e.g., strength and power)

*The actual training goals and priorities are based on the specific sport or activity and may differ from the goals listed here.

Postseason (active rest)	Variable	Variable	Not specific (may include activities other than sport skill or resistance training)
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*The actual training goals and priorities are based on the specific sport or activity and may differ from the goals listed here.

Step 2: Exercise Selection

Exercise selection involves choosing exercises for a resistance training program. To make informed exercise selections, the strength and conditioning professional must understand the nature of various types of resistance training exercises, the movement and muscular requirements of the sport, the athlete's exercise technique experience, the equipment available, and the amount of training time available. [Figure 18.2](#) on [page 528](#) displays exercise selection flow charts for lower body strength and power exercises.

Exercise Type

Although there are hundreds of resistance training exercises to select from when one is designing a program, most involve primary muscle groups or body areas and fall into categories based on their relative importance to the athlete's sport.

Core and Assistance Exercises

Exercises can be classified as either core or assistance based on the size of the muscle areas involved and their level of contribution to a particular sport movement. **Core exercises** recruit one or more large muscle areas (i.e., chest, shoulder, back, hip, or thigh), involve two or more primary joints (**multijoint exercises**; e.g., split squat, bench press, bent-over row), and receive priority when one is selecting exercises because of their direct application to the sport and enhanced dynamic correspondence (i.e., transference of gym-based training to improved performance in athletic tasks). **Assistance exercises** usually recruit smaller muscle areas (i.e., upper arm, abdominal muscles, calf, neck, forearm, lower back, or anterior lower leg), involve only one primary joint (**single-joint exercises**; e.g., knee extension and flexion, biceps curls), and are considered less important for improving sport performance. Generally, all the joints at the shoulder—the glenohumeral and shoulder girdle articulations—are considered one *primary* joint when resistance training exercises are categorized as core or assistance. The spine is similarly considered a single primary joint (e.g., abdominal crunch and back extension exercises).

Application of the Needs Analysis

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A	Scenario B	Scenario C
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The information in this table reflects one approach to evaluating the requirements of a sport and profiling an athlete.

Female collegiate basketball player In-season	Male professional American football lineman Off-season	Male high school cross country runner Preseason
SPORT EVALUATION Movement analysis <i>Sport:</i> Running and jumping, ball handling, shooting, blocking, and rebounding <i>Muscular involvement:</i> All major muscle areas, especially the hips, thighs, and shoulders Physiological analysis (primary requirement) Strength/power	SPORT EVALUATION Movement analysis <i>Sport:</i> Grabbing, pushing, repelling, and deflecting opponents <i>Muscular involvement:</i> All major muscle areas, especially the hips, thighs, chest, arms, and low back Physiological analysis (primary requirement) Hypertrophy	SPORT EVALUATION Movement analysis <i>Sport:</i> Running, repetitive leg and arm movements <i>Muscular involvement:</i> All lower body muscle areas, postural muscles, shoulders, and arms Physiological analysis (primary requirement) Muscular endurance
ATHLETE'S PROFILE Training background - Has resistance trained regularly since high school - Possesses excellent skill in performing free weight and machine exercises - Just completed a 3×/week resistance training program in the preseason consisting of full-body workouts that included lower and upper body exercises during each session (4 core, 3 sets of 3 reps using loads ranging from 4RM-6RM)	ATHLETE'S PROFILE Training background - Has resistance trained regularly throughout high school, college, and his professional career - Possesses excellent skill in performing free weight and machine exercises - Just completed a 2×/week resistance training program in the postseason^b consisting of 8 exercises (3 core, 5 assistance; 4 lower body, 4 upper body, 2-3 sets of 12RM-15RM loads)	ATHLETE'S PROFILE Training background - Just began resistance training in preseason - Has only limited skill in performing free weight and machine exercises - Just completed a 3×/week resistance training program in the off-season^c consisting of full-body workouts that included lower and upper body exercises during each session (2-3 sets of 6-8 reps using loads ranging from 8RM-10RM)
CLASSIFICATION OF RESISTANCE TRAINING STATUS Advanced	CLASSIFICATION OF RESISTANCE TRAINING STATUS Advanced	CLASSIFICATION OF RESISTANCE TRAINING STATUS Beginner
PRIMARY IN-SEASON RESISTANCE TRAINING GOAL Strength/power ^a	PRIMARY OFF-SEASON RESISTANCE TRAINING GOAL Hypertrophy	PRIMARY PRESEASON RESISTANCE TRAINING GOAL Muscular endurance

The information in this table reflects one approach to evaluating the requirements of a sport and profiling an athlete.

COMMENTS ^aThe in-season will address both of these goals through a combination of appropriate exercise selection and volume-load assignments.	COMMENTS ^bDue to the extreme physical demands of American football, this athlete's postseason training volume was greater than is often assigned for the active rest phase of a typical program.	COMMENTS ^cBecause this athlete just began his resistance training program, his frequency was limited to 3×/week in the off-season rather than the 4×/week that are typically completed by better-trained individuals.
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The information in this table reflects one approach to evaluating the requirements of a sport and profiling an athlete.

A common application of assistance exercises is for injury prevention and rehabilitation, because these exercises often isolate a specific muscle or muscle group. The muscles that are predisposed to injury from the unique demands of a sport skill (e.g., the shoulder external rotators for overhand pitching) or those that require reconditioning after an injury (e.g., a quadriceps contusion) can be specifically conditioned by an assistance exercise. However, it is important that the chosen exercises improve a physical characteristic within the program (flexibility, strength, etc.) and that the professional prescribes them appropriately with specific volumes and intensities.

Structural and Power Exercises

A core exercise that emphasizes loading the spine directly (e.g., back squat) or indirectly (e.g., power clean) can be further described as a **structural exercise**. More specifically, a structural exercise involves muscular stabilization of posture during performance of the lifting movement (e.g., maintaining a rigid torso and a neutral spine during the back squat). Practitioners should be mindful that during traditional structural exercises (e.g., squat and bench press), a substantial portion of the range of motion results in deceleration (47, 93, 119). As such, ballistic tasks that emphasize movement velocity (e.g., loaded jumps) should be programmed to emphasize power development. A structural exercise that is performed very quickly or explosively is considered a **power exercise**. Typically, power exercises are assigned to athletes when they are appropriate for the athlete's sport-specific training priorities. It should be noted, however, that in weak athletes, or those with a low training age, emphasizing strength development has been shown to enhance power output to an equal or greater extent than ballistic training (26, 28, 29). Semi-ballistic exercises, such as weightlifting exercises and their derivatives, can also be used effectively for the development of power because they provide the potential to emphasize a velocity- or force-dominant stimulus ([figure 18.6](#)) (21, 170-173). Such a diverse range of exercises also permits substantial variation and the opportunity to

select exercises to accommodate technique, mobility, and injury-related consideration.

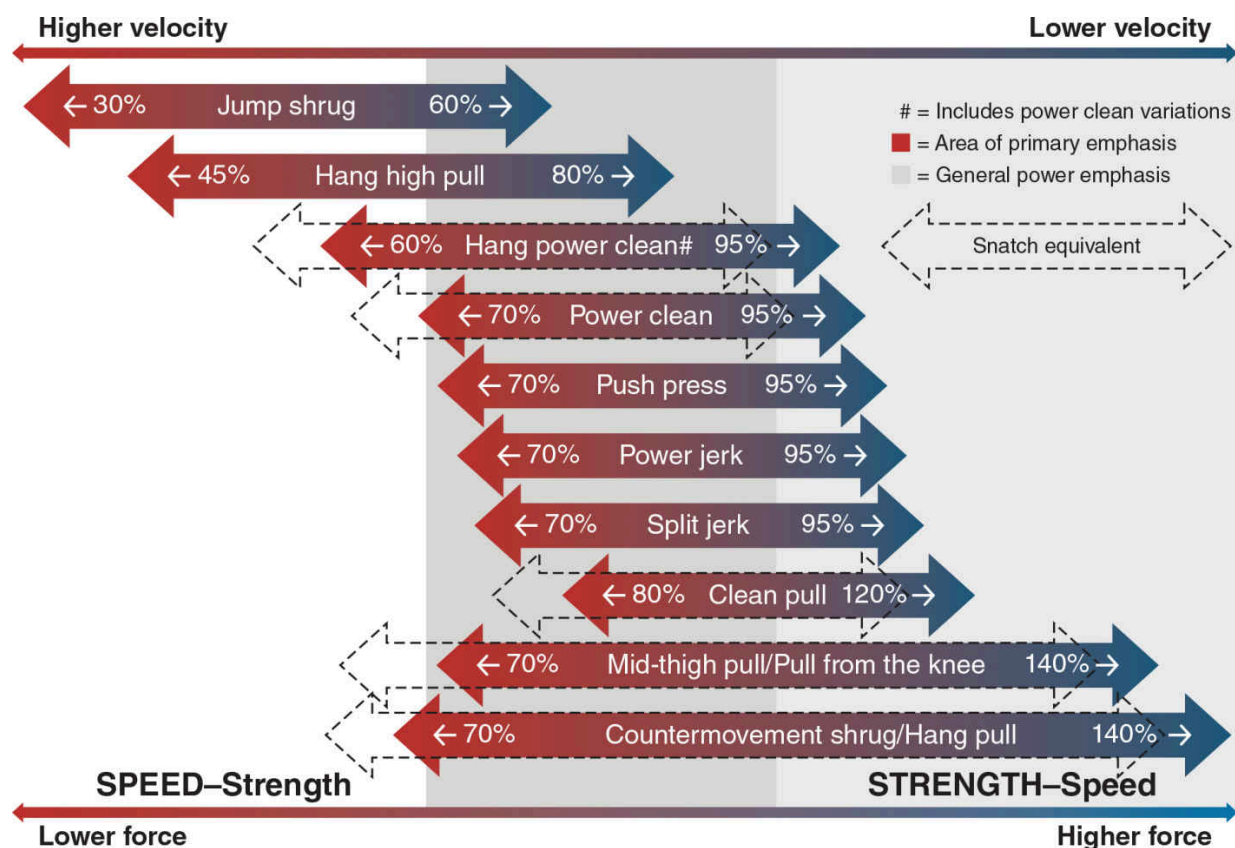


FIGURE 18.6 Recommended loading ranges to emphasize speed–strength, strength–speed, or general power development using weightlifting exercises and their derivatives. Red shading illustrates the primary loading emphasis for the exercises, whereas blue shading illustrates loading, which is likely to be less effective at enhancing the specific strength characteristic. Loads are based on the 1RM for the power clean, power snatch, or the overhead derivative, as appropriate.

Adapted by permission from P. Comfort, G.G. Haff, T.J. Suchomel, et al., “National Strength and Conditioning Association Position Statement on Weightlifting for Sports Performance,” *Journal of Strength and Conditioning Research* 37, no. 6 (2023): 1168.

Movement Analysis of the Sport

In the needs analysis (step 1), the strength and conditioning professional has identified the unique requirements and characteristics of the sport. The exercises selected for a resistance training program that focus on conditioning for a particular sport need to be relevant to the activities of that sport in their body and limb movement patterns, joint ranges of motion, and muscles involved and muscle actions. The exercises should also create muscular balance to reduce risk of injury from disproportionate training. For example, the hamstrings serve as knee flexors and hip extensors, while the quadriceps are knee extensors. Researchers have shown that the muscle activation of the hamstrings and

quadriceps may vary based on the exercises that are chosen (101, 153, 190). Thus, it is up to the strength and conditioning professional to understand how each exercise may affect muscular development and to prescribe exercises accordingly based on the needs of his or her athletes.

Sport-Specific Exercises

The more similar the training activity is to the actual sport movement, the greater the likelihood that there will be a positive transfer to that sport (i.e., dynamic correspondence) (160, 165). This is the specificity concept, sometimes confused with the SAID principle. Arguably the SAID principle is more important than movement specificity; strength and conditioning professionals need to be mindful of the specific physiological adaptations that are targeted during each phase of training. [Table 18.3](#) provides examples of resistance training exercises that relate in varying degrees to the movement patterns of various sports. The strength and conditioning professional should find this table helpful when trying to identify sport-specific exercises to ensure dynamic correspondence. For example, the primary muscles involved in jumping for basketball are the hip and knee extensors. An athlete can train these muscles by performing the leg press or back squat, but which exercise is preferable? Certainly, both exercises strengthen the hip and knee extensors, but because jumping is performed from an erect body position with balance and weight-bearing forces as considerations, the back squat is more relevant to jumping and is therefore preferred over the leg press (12, 84). Additionally, full hip extension is achieved during a squat and a jump, but this is not the case for all leg press machines. The weightlifting movements and their derivatives are relevant to jumping because of their quick movement characteristics, thereby applying a high rate of force development and power output (16, 87).

TABLE 18.3 Examples of Movement-Related Resistance Training Exercises

Movement pattern	Related exercises
Ball dribbling and passing	Closed-grip bench press, dumbbell bench press, triceps pushdown, reverse curl, hammer curl
Ball kicking	Unilateral hip adduction and abduction, single-leg squat, forward step lunge, leg (knee) extension, leg raise
Freestyle swimming (including start and turns)	Pull-up, lateral shoulder raise, forward step lunge, upright row, barbell pullover, single-leg squat
Vertical jumping (ballistic, slow SSC)	Snatch, power clean, push jerk, back squat, front squat, loaded squat jump, jump shrug, push press, push jerk

Vertical jumping (plyometric, fast SSC)	Drop jump, rebound jump, standing calf raise, seated calf raise
Racket stroke	Flat dumbbell fly, lunge, bent-over lateral raise, wrist curl, wrist extension
Rowing	Power clean, clean pull, snatch pull, hang high pull, bent-over row, seated row, back squat, leg press, deadlift, stiff-leg deadlift, good morning
Running, sprinting	Snatch, clean, power clean, back squat, front squat, forward step lunge, step-up, leg (knee) extension, leg (knee) curl, toe raise (dorsiflexion)
Throwing, pitching	Lunge, single-leg squat, barbell pullover, overhead triceps extension, shoulder internal and external rotation

Muscle Balance

Exercises selected for the specific demands of the sport should maintain a balance of muscular strength across joints and between opposing muscle groups (e.g., biceps brachii and triceps brachii). Avoid designing a resistance training program that increases the risk of injury due to a disparity between the strength of the **agonist**, the muscle or muscle group actively causing the movement (e.g., the quadriceps in the leg [knee] extension exercise), and the **antagonist**, the sometimes passive (i.e., not concentrically involved) muscle or muscle group located on the opposite side of the limb (e.g., the hamstrings in the leg [knee] extension exercise). If an imbalance is created or discovered, exercises to restore an appropriate strength balance need to be selected. For example, if isokinetic testing reveals that the hamstrings are extremely weak compared with the quadriceps, additional hamstring exercises could be included to compensate for the imbalance (48). Isokinetic adaptations may be specific to the muscle actions involved as well as the speeds used during the training protocols (31, 129) and do not transfer effectively to enhanced performance in athletic tasks (94, 122). Note that **muscle balance** does not always mean equal strength, just a proper ratio of strength, power, or muscular endurance of one muscle or muscle group relative to another muscle or muscle group.

Exercise Technique Experience

An important part of the needs analysis described earlier is evaluating the athlete's training status and exercise technique. If there is any question whether an athlete can perform an exercise with proper technique, the strength and conditioning professional should ask the athlete to demonstrate the exercise. If the athlete uses incorrect technique, the strength and conditioning professional should provide complete instruction. Often, unskilled individuals are introduced to machines and free weight assistance exercises because these are considered easier to perform than free weight core exercises due to their lower balance and coordination requirements (146, 147). Despite this, one should not assume that

the athlete will perform exercises correctly, even those that are relatively easy to perform. Thus, a strength and conditioning professional should provide appropriate exercise progressions and regressions that are based on each athlete's technique and their ability to maintain their technique with heavier loads. An example of squat progression and regression exercises is displayed in [figure 18.7](#).

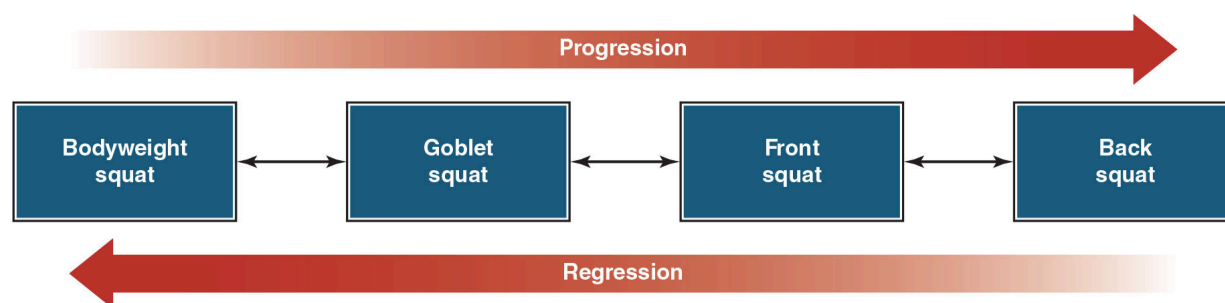


FIGURE 18.7 Squat progression and regression exercise examples. *Note:* Individual exercises may also be progressed or regressed using different loads. In addition, advanced athletes may progress the front squat and back squat using variable resistance training with chains or bands as well as accentuated eccentric loading.

Availability of Resistance Training Equipment

The availability of training equipment must be considered in the selection of exercises. A lack of certain equipment may necessitate selecting exercises that are not as sport specific. For example, the absence of Olympic-type barbells with revolving sleeves would preclude exercises such as the power clean, and an insufficient supply of barbell plates may result in substituting exercises that do not require as much resistance; for example, the back squat could be replaced by the front squat.

Available Training Time Per Session

The strength and conditioning professional should weigh the value of certain exercises against the time it takes to perform them. Some exercises take longer to complete than others. If time for a training session is limited, exercises that are more time efficient may need to be given priority over others. For example, the machine leg press could be selected instead of the free weight lunge to train the hips and thighs of a 100-meter sprinter. The time required to move the machine pin to the correct slot in a weight stack and perform 5 repetitions of a machine leg press is much less than the time required for the lunge exercise, for which the athlete must load both ends of a bar, attach the locks, back out of the power rack, establish a stable starting position, perform 5 repetitions of *each* leg, and rerack the bar. Although the machine leg press is less sport specific, the

time saved may permit including other exercises or performing more sets. The benefit of including the more specific lunge exercise, on the other hand, may be worth the additional time needed, although this depends on the goals of the training session and time available.

Application of the Exercise Selection Guidelines

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes. Exercises are not listed in order of execution.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
CORE Push jerk (total body, power) ^a Hang power clean (total body, power) ^a Jump shrug (total body, power) ^a Front squat (hip and thigh) Romanian deadlift (posterior hip and thigh) Incline bench press (chest) Seated row (upper back) Pull-up (back, shoulders, arms)	CORE^b Power clean (total body, power) Tire flipping (total body, power) Back squat (hip and thigh) Deadlift (hip and thigh) Bench press (chest) Shoulder press (shoulders)	CORE Hexagonal barbell deadlift (hip and thigh) Lunge (hip and thigh) ^c Rear leg–elevated split-squat (hip and thigh) Single-leg stiff-legged deadlift (posterior hip and thigh) Vertical chest press (chest) ^d
ASSISTANCE Abdominal crunch (abdomen) Single-leg stiff-legged deadlift (posterior hip and thigh) Standing calf raise (posterior lower leg)	ASSISTANCE Towel-grip pull-up (forearm grip) Abdominal crunch (abdomen) Step-up (hip and thigh) Leg (knee) curl (posterior thigh) Bent-over row (upper back) Shoulder shrug (upper back and neck) Barbell biceps curl (anterior upper arm) Lying triceps extension (posterior upper arm) Seated calf (heel) raise (posterior lower leg)	ASSISTANCE Abdominal crunch (abdomen) Leg (knee) curl (posterior thigh) Lateral shoulder raise (shoulders) One-arm dumbbell row (upper back) ^e Toe raise (dorsiflexion) (anterior lower leg) Machine back extension (lower back) Cable hip flexion (hip flexors)
COMMENTS	COMMENTS	COMMENTS

^aThese exercises are included to maximize power and match the jumping movements of basketball.

This athlete has extra time available to perform more resistance training exercises because sport skill practice is not the first priority in the off-season.

^bGreater training frequency allows more core exercises to be included (see step 3).

^cAlthough not always considered a core exercise, the lunge recruits muscles and joints that have direct application to running.

^dThis exercise also involves the triceps brachii muscle, so an assistance exercise to isolate the triceps is not needed. This reduces the time devoted to the resistance training portion of the preseason program.

^eThis exercise also involves the biceps brachii muscle, so an assistance exercise to isolate the biceps is not needed. This reduces the time devoted to the resistance training portion of the preseason program.

Step 3: Training Frequency

Training frequency refers to the number of training sessions completed in a given time period. For a resistance training program, a common time period is one week. When determining training frequency, the strength and conditioning professional should consider the athlete's training status, sport season, projected exercise loads, types of exercises, and other concurrent training or activities. When the weekly resistance training volume-load is equated, hypertrophic adaptations (142) and strength adaptations (34) are comparable irrespective of training frequency.

Training Status

The athlete's level of preparedness for training, which was determined during the needs analysis (step 1), is an influential factor in determining training frequency because it affects the number of rest days needed between training sessions. Traditionally, three workouts per week are recommended for many athletes, because the intervening days allow sufficient recovery between sessions. As an athlete adapts to training and becomes better conditioned, it is appropriate to consider increasing the number of training days to four and, with additional training, maybe five, six, or seven ([table 18.4](#)), although this depends on the phase of the competitive cycle. The general guideline is to schedule training sessions to include at least one rest or recovery day—but not more than three—between sessions that stress the same muscle groups (77); however, this may also depend on the intensity of the prescribed exercises. For example, if a strength and conditioning professional wants a novice athlete to perform a total body resistance training program two times per week, the sessions should be spaced out evenly (e.g., Monday and Thursday or Tuesday and Friday). If the athlete trains only on Monday and Wednesday, the absence of a training stimulus between Wednesday and the following Monday may result in a *decrease* in the athlete's training status (41, 61, 77), although, for a short time in well-trained athletes, one session per week can maintain strength if appropriate loading strategies are implemented (41, 61). Examples of training frequencies of 3 and 4 times per week using a total body push-pull format are presented in [table 18.5](#).

TABLE 18.4 Resistance Training Frequency Based on Training Status

Training status	Frequency guidelines (sessions per week)
Beginner	2-3
Intermediate	3-4
Advanced*	4-7

*May include multiple sessions per day.

Data from Graves et al. (1988); Häkkinen et al. (1988); Hoffman et al. (1990).

TABLE 18.5 Examples of Common Total-Body Routines Using a Push–Pull Emphasis

Training organization	Body parts or muscle groups trained	Sample training week							Resulting training frequency
		Sun	Mon	Tues	Wed	Thurs	Fri	Sat	
Push–pull	Total body	Rest	Heavy push	Rest	Heavy pull	Rest	Light mix	Rest	3 times per week
Push–pull	Total body	Rest	Heavy push	Rest	Heavy pull	Light push	Rest	Light pull	4 times per week

More highly resistance-trained (intermediate or advanced) athletes can augment their training by using a **split routine** in which different muscle groups are trained on different days. Training nearly every day may seem to violate the recommended guidelines for recovery, but grouping exercises that train a portion of the body (e.g., lower body or upper body) or certain muscle areas (e.g., chest, shoulder, and triceps) gives the trained athlete an opportunity to adequately recover between similar training sessions ([table 18.6](#)). For instance, a common lower body–upper body regimen includes four training sessions per week: lower body on Monday and Thursday and upper body on Tuesday and Friday (or vice versa). This way, there are two or three days of rest between each lower or upper body training session, even though the athlete trains on two consecutive days twice a week (80). For split routines with three distinct training days, the rest days are not on the same day each week. It should be noted that there are no differences in muscle hypertrophy or strength adaptations between split and total body routines when training volume is equated (131). However, since volumes may not always be matched in training, it is possible that a split routine may allow athletes to accumulate a greater volume–load per muscle group, potentially benefiting hypertrophy and general strength characteristics. Given that athletes may have limited time to train, and the fact that sports do not isolate specific muscle groups, strength and conditioning professionals must determine whether a split routine is appropriate for their athletes and when and how it could be implemented. For example, an

off-season training program may be an appropriate time to implement a split routine because this is typically the time when athletes may focus on developing muscle hypertrophy, which has been associated with higher training volumes (143). In contrast, preseason and in-season phases of training may not be appropriate for split routines given the emphasis on developing maximal and rapid force production characteristics, reducing training volume and fatigue, and improving sport-specific technical skills.

TABLE 18.6 Examples of Common Split Routines

Training day	Body parts or muscle groups trained	Sample training week							Resulting training frequency
		Sun	Mon	Tues	Wed	Thurs	Fri	Sat	
1	Lower body	Rest	Lower body	Upper body	Rest	Lower body	Upper body	Rest	4 times per week
2	Upper body								
1	Chest, shoulders, triceps	Rest	Chest, shoulders, triceps	Lower body	Back, trapezius, biceps	Rest	Chest, shoulders, triceps	Lower body	5 times per week*
2	Lower body								
3	Back, trapezius, biceps								
1	Chest and back	Chest and back	Lower body	Shoulders and arms	Rest	Chest and back	Lower body	Shoulders and arms	6 times per week*
2	Lower body								
3	Shoulders and arms								

*Frequency varies between five times per week and six times per week, depending on the day of the week that is the first training day.

Sport Season

Another factor that influences resistance training frequency is the sport season. For example, the increased emphasis on practicing the sport skill during the in-season necessitates a decrease in the time spent in the weight room and, consequently, reduces the frequency of resistance training ([tables 18.4 and 18.7](#)). The problem is that there simply is not enough time to fit all the desired

modes of training into each day. So even though a well-trained athlete may be capable of completing four or more resistance training sessions per week, the other time demands of the sport (i.e., competition schedule, technical and tactical training) may not permit this. When time is constrained, the subdivision of weekly training volume across multiple short-duration bouts, termed **microdosing**, has been suggested as a potentially useful strategy, which may mitigate the fatigue associated with longer duration, higher volume training sessions (34, 35).

TABLE 18.7 Resistance Training Frequency Based on the Sport Season (for a Trained Athlete)

Sport season	Frequency guidelines (sessions per week)
Off-season	4-6
Preseason	3-4
In-season	1-3
Postseason (active rest)	0-3

Data from Stone and O’Bryant (1987); Tesch (1992).

Training Load and Exercise Type

The ability to train more frequently may be enhanced by alternating heavier and lighter training days (42, 166, 177). As shown in [table 18.5](#), heavy and light training days may be organized based on the type of exercises that are being performed regardless of whether the same muscle groups are being used. While training to failure may lengthen the necessary recovery time between sessions (59, 117), athletes may not require longer recovery times despite training with heavier loads if the volume remains low for the selected exercises (5). However, strength and conditioning professionals should be aware that higher volume loads can lengthen the necessary recovery time between training sessions (124). There is also evidence that upper body muscles can recover more quickly from heavy loading sessions compared to lower body muscles (76). The same is true regarding an athlete’s ability to recover faster from single-joint exercises compared to multijoint exercises (158). These research findings may explain why, for example, powerlifters may schedule only one very heavy deadlift or squat training session per week.

More importantly, in a recent meta-analysis, Swinton and colleagues (182) concluded that the most important factor for appropriate adaptations is the intensity used, with heavy loads ($\geq 80\%$ 1RM) resulting in the greatest increases in strength and with low to moderate loads ($\leq 30\%$ 1RM) during ballistic tasks

stimulating the greatest increases in vertical jump performance and moderate loads (40%-70% 1RM) being the most appropriate for power development. However, as mentioned previously, an emphasis on strength development in weaker individuals may result in the greatest improvements in performance (8, 26, 29, 30, 178). This is likely due to the fact that relative force production determines acceleration, with the duration over which the force is applied determining acceleration, based on the impulse–momentum theorem.

Other Training

Exercise frequency is also influenced by the overall amount of physical stress, so the strength and conditioning professional must consider the effects of all forms of exercise. If the athlete's program already includes aerobic endurance (e.g., long, slow distance) or speed, agility, and power (e.g., sprinting, agility, speed–endurance, plyometric) training, sport skill practice, or any combination of these components, the frequency of resistance training may need to be reduced (32). Additionally, the effects of a physically demanding occupation may be relevant. Athletes who work in manual labor jobs, instruct or assist others in physical activities, or are on their feet all day may not be able to withstand the same training frequency as athletes who are less active outside of their sport-related pursuits.

Step 4: Exercise Order

Exercise order refers to a sequence of resistance exercises performed during one training session. Although there are many ways to arrange exercises, decisions are invariably based on how one exercise affects the quality of effort or the technique of another exercise. Usually, exercises are arranged so that an athlete's maximal force capabilities are available (from a sufficient rest or recovery period) to complete a set with proper exercise technique. Four of the most common methods of ordering resistance exercises are described in the following paragraphs.

Application of the Training Frequency Guidelines

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
ADVANCED TRAINING STATUS ALLOWS 4-7×/week	ADVANCED TRAINING STATUS ALLOWS 4-7×/week	BEGINNER TRAINING STATUS ALLOWS 2-3×/week
FREQUENCY GUIDELINE BASED ON THE SPORT SEASON 1-3×/week	FREQUENCY GUIDELINE BASED ON THE SPORT SEASON 4-6×/week	FREQUENCY GUIDELINE BASED ON THE SPORT SEASON 3-4×/week
ASSIGNED RESISTANCE TRAINING FREQUENCY 3×/week ^a Total body workouts performed on Monday, Wednesday, and Friday while incorporating a push, pull, and mix emphasis	ASSIGNED RESISTANCE TRAINING FREQUENCY 4×/week (split routine ^b) Monday and Thursday (lower body exercises) Tuesday and Friday (upper body exercises)	ASSIGNED RESISTANCE TRAINING FREQUENCY 2×/week ^c Total body workouts performed on Wednesday and Saturday while incorporating a heavy and light emphasis
COMMENTS ^a Training frequency is maintained from the previous season (preseason) to allow for the maintenance and improvement of strength–power characteristics during the in-season phase of training.	COMMENTS ^b A split routine allows for more overall exercises to be performed without an excessive increase in training time (per session) because the exercises are divided over more training days.	COMMENTS ^c Despite the sport season, the assigned training frequency is based on the athlete's training status and the additional need to develop cardiovascular endurance.

Power, Other Core, Then Assistance Exercises

Power exercises such as weightlifting movements and their derivatives (e.g., snatch, hang power clean, jump shrug) and other ballistic exercises (e.g., jump squat or hexagonal barbell jump) should be performed first in a training session, followed by other non-power core exercises, and then assistance exercises. This is due to the large neuromuscular demand placed on the body by power exercises because they require greater levels of skill and are prone to fatigue (71). The literature also refers to this arrangement as *multijoint exercises and then single-joint exercises* or *large muscle areas and then small muscle areas* (151, 183). Athletes who become fatigued are prone to using poor technique and consequently are at higher risk of injury. The explosive movements and extensive muscular involvement of power exercises also result in substantial energy expenditure (138). This is another reason to have athletes perform such exercises first, while they are still metabolically fresh. If power exercises are not selected in step 2 (exercise selection), the recommended order of exercises is core exercises (ideally with the most demanding exercises first) and then assistance exercises. This is an important consideration because researchers have shown that greater maximal strength adaptations may be produced with the exercises programmed earlier in a training session (44, 152) while the latter exercises may be negatively affected due to the accumulated fatigue (116, 150, 152).

Lower and Upper Body Exercises (Alternated)

One method of providing the opportunity for athletes to recover more fully between exercises is to alternate lower body exercises with upper body exercises. This arrangement is especially helpful for untrained individuals who find that completing several lower or upper body exercises in succession is too strenuous (50). Also, if training time is limited, this method of arranging exercises minimizes the length of the rest periods required between exercises and maximizes the rest between body areas. The result is a decrease in overall training time because the athlete can perform a lower body exercise and then immediately go to an upper body exercise without having to wait for the lower body to rest. However, when choosing an order between lower and upper body exercises, strength and conditioning professionals should consider prescribing lower body exercises first since these movements typically involve more muscle mass and may place a greater overall stress on the body due to the absolute loads that will be used. If the exercises are performed with minimal rest periods (20-30 s), this method is also referred to as **circuit training** (generally not optimal for the development of maximal strength)—a method sometimes also

used to improve cardiovascular endurance (58, 130), although to a lesser extent than conventional aerobic exercise training.

“Push” and “Pull” Exercises (Alternated)

Another method of improving efficiency of training when time is limited is to alternate pushing exercises (e.g., bench press, shoulder press, triceps extension) with pulling exercises (e.g., lat pulldown, bent-over row, biceps curl) (2). This push–pull arrangement ensures that the same muscle group will not be used in two exercises (or sets, in some cases) in succession, thus reducing fatigue in the involved muscles. This approach is similar to the programming of antagonistic supersets, described later. In contrast, arranging several pulling exercises (e.g., pull-up, seated row, hammer curl) one after the other, even with a rest period between each, will compromise the number of repetitions performed because the biceps brachii muscle (involved in all three exercises) will become less responsive due to fatigue. The same result would occur if several pushing exercises (e.g., incline bench press, shoulder press, triceps pushdown) were sequentially arranged (all three engage the triceps brachii) (150). There are also push–pull arrangements for the lower body—for example, back squat and front squat as “push” and stiff-legged deadlift and leg (knee) curl as “pull”—but the classification of some exercises as “push” or “pull” is not as clear (e.g., leg [knee] extension). However, in such examples strength and conditioning professionals need to be mindful when selecting such exercises to ensure that key stabilizing muscles (e.g., erector spinae) are not excessively fatigued in both exercises. Alternating push and pull exercises is also used in circuit training programs and is an ideal arrangement for novice athletes or those returning to a resistance training program (3).

Supersets and Compound Sets

Other methods of arranging exercises involve having athletes perform one set of a pair of exercises with little to no rest between them. Two common examples are supersets and compound sets. A **superset** involves two sequentially performed exercises that stress two opposing muscles or muscle areas (i.e., an agonist and its antagonist) (2), similar to the push–pull approach. For example, an athlete performs 10 repetitions of the barbell biceps curl exercise, sets the barbell down, then goes over to the triceps pushdown station and performs 10 repetitions. A **compound set** involves sequentially performing two different exercises for the same muscle group (2). For instance, an athlete completes a set of the barbell biceps curl exercise, then switches to dumbbells and immediately performs a set of the hammer curl exercise. In this case, the stress on the same muscle is compounded because both exercises recruit the same

muscle area. Both methods of arranging and performing pairs of exercises are time efficient and purposely more demanding—and consequently may not be appropriate for unconditioned athletes. Note, however, that sometimes the meanings of *superset* and *compound set* are interchanged.

Application of the Exercise Order Guidelines

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
ASSIGNED EXERCISE ORDER STRATEGIES Power, other core, then assistance exercises “Push” and “pull” exercises (alternated)	ASSIGNED EXERCISE ORDER STRATEGIES Core and then assistance exercises “Push” and “pull” exercises (alternated)	ASSIGNED EXERCISE ORDER STRATEGIES Core and then assistance exercises Lower and upper body exercises (alternated)
MONDAY Push jerk Front squat ^a Incline bench press ^b Triceps pushdown WEDNESDAY Hang power clean Romanian deadlift ^a Seated row ^b Dumbbell alternating curl FRIDAY Jump shrug Rear leg–elevated split-squat ^a Pull-up ^b Single-leg stiff-legged deadlift	LOWER BODY (MONDAY AND THURSDAY) Deadlift ^c Back squat ^c Step-up ^c Leg (knee) curl Seated calf (heel) raise UPPER BODY (TUESDAY AND FRIDAY) Bench press Bent-over row Shoulder press Barbell biceps curl ^d Shoulder shrug Lying triceps extension Abdominal crunch	WEDNESDAY Hexagonal barbell deadlift Vertical chest press One-arm dumbbell row Abdominal crunch Machine back extension ^e SATURDAY Lunge Lateral shoulder raise Single-leg stiff-legged deadlift Toe raise (dorsiflexion) Abdominal crunch Machine back extension ^e
COMMENTS ^{a,b} These exercises are alternated to provide relative rest between their similar movement patterns while still following the “power, other core, then assistance” exercise order strategy.	COMMENTS ^c These exercises do not follow the “push” and “pull” (alternated) exercise arrangement and could be performed in a variety of sequences (e.g., back squat, deadlift, step-up).	COMMENTS ^e Exercises that concentrically train the lower back muscles should be performed after exercises that require an erect torso or a neutral spine position (e.g., lunge and lateral shoulder raise). Fatigue of the lower back muscles can result in incorrect and potentially injurious exercise